Project Report: Thass Chat

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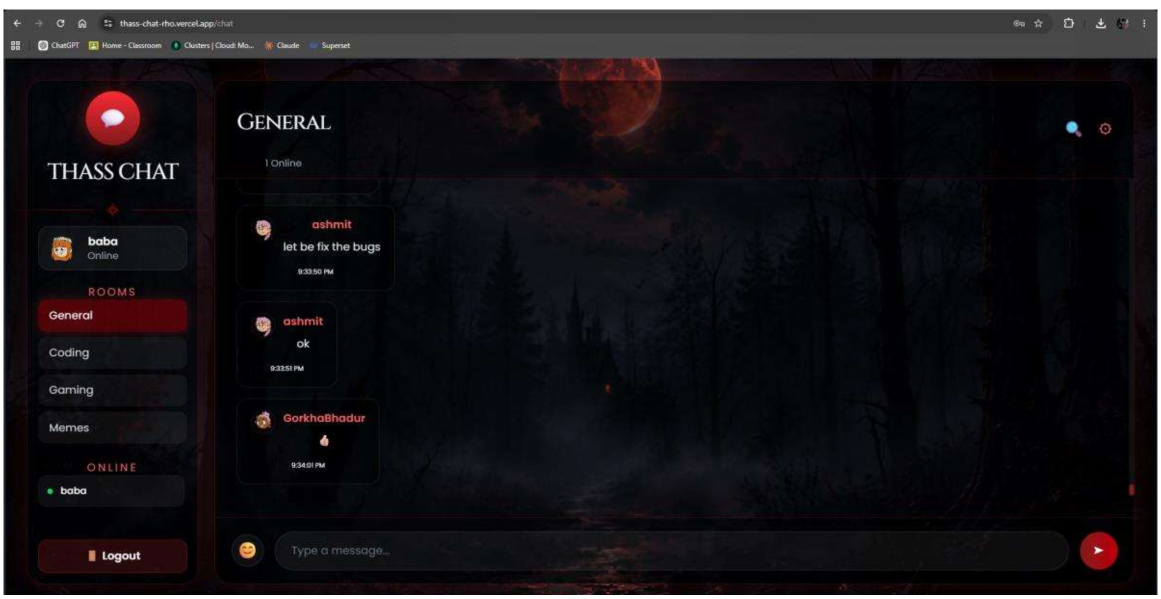
1. Abstract

Thass Chat is a real-time multi-room chat application built using the MERN ecosystem with Socket.IO for instant messaging. The system provides secure authentication, persistent message storage, room-based communication, typing indicators, online user tracking, responsive design, and cloud deployment using Vercel and Render.



2. Objectives

- Develop a secure authentication system using JWT.
- Enable real-time communication using Socket.IO.
- Provide multiple chat rooms.
- Store chat history in MongoDB Atlas.
- Deploy the application on cloud platforms.
- Create a responsive and modern UI.

3. Technology Stack

Frontend: React (Vite), Tailwind CSS, Framer Motion
Backend: Node.js, Express.js, Socket.IO
Database: MongoDB Atlas
Media: Cloudinary
Authentication: JWT, bcrypt
Deployment: Vercel and Render
Version Control: Git & GitHub

4. Features

- User registration and login
- JWT authentication
- Real-time messaging
- Multiple chat rooms
- Typing indicator
- Online user tracking
- Persistent message history
- Responsive UI
- Glassmorphism dark theme

5. System Architecture

Client (React) communicates with the Express backend through REST APIs and Socket.IO. The backend authenticates users, stores data in MongoDB Atlas, uploads media to Cloudinary, and broadcasts messages through Socket.IO.

6. Challenges

- Managing Socket.IO room synchronization.
- Handling deployment and CORS.
- Synchronizing online users.
- Designing a responsive interface.

7. Future Scope

- Direct messaging
- File sharing
- Voice/video calls
- Push notifications
- Read receipts
- Message search

8. Conclusion

Thass Chat demonstrates a complete full-stack web application integrating secure authentication, real-time communication, cloud deployment, and modern UI design. The project showcases practical software engineering concepts and provides a strong foundation for future enhancements.